

1. Sheffer stroke (9 points)

Is the Sheffer stroke $|$ associative? Show this by proving the semantic equivalence $(\phi | \psi) | \chi \equiv \phi | (\psi | \chi)$, or bring a counterexample to disprove it.

SOLUTION

ϕ	ψ	χ	$(\phi \psi)$	$(\psi \chi)$	$(\phi \psi) \chi$	$\phi (\psi \chi)$
T	T	T	F	F	T	T
T	T	F	F	T	T	F
T	F	T	T	T	F	F
T	F	F	T	T	T	T
F	T	T	T	F	F	T
F	T	F	T	T	T	T
F	F	T	T	T	F	T
F	F	F	T	T	T	T

As the last two columns of the truth table do not match on every row, the Sheffer stroke is not associative. A counterexample is when ϕ and ψ are true and χ is false: in this case, $(\phi | \psi) | \chi$ is true, but $\phi | (\psi | \chi)$ is false.

2. Island puzzle (12 points)

On the island of liars and truth speakers, every resident either always speaks the truth or always lies. Suppose that you meet three islanders, A, B and C. A says '*B is a liar*', B says '*C is a truth speaker*', and C says '*A and B are both liars*'. Determine, via a truth table, whether A, B and C are liars or truth speakers.

SOLUTION:

Let us use three propositional variables:

t_A : A is a truth speaker

t_B : B is a truth speaker

t_C : C is a truth speaker

We can construct the following truth table:

t_A	t_B	t_C	$t_A \leftrightarrow \neg t_B$	$t_B \leftrightarrow t_C$	$t_C \leftrightarrow \neg t_A \wedge \neg t_B$
T	T	T	F	T	F
T	T	F	F	F	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	T	F
F	T	F	T	F	T
F	F	T	F	F	T
F	F	F	F	T	F

The highlighted row is the only one which makes all three bi-implications true, which means it is the only one which can represent reality. Therefore, A must be a truth speaker, while B and C must be liars.

3. CNF algorithm (12 points)

Show, using the CNF algorithm, that the formula $(p \wedge \neg q) \rightarrow \neg(q \rightarrow p)$ is semantically equivalent to $\neg p \vee q$.

SOLUTION:

1. Implication-free:
 $\neg(p \wedge \neg q) \vee \neg(q \rightarrow p)$
2. Implication-free:
 $\neg(p \wedge \neg q) \vee \neg(\neg q \vee p)$
3. Bringing negations inside brackets:
 $(\neg p \vee q) \vee (q \wedge \neg p)$
4. Distributivity:
 $(\neg p \vee q \vee q) \wedge (\neg p \vee q \vee \neg p)$
5. Simplifying the solution:
 $\neg p \vee q$

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg q \vee r)$ to show whether it is satisfiable.

SOLUTION

1. No unit clauses to eliminate.
2. Pure literal: r. Let us try $r \equiv T$:
 $(p \vee \neg q) \wedge (\neg p \vee q) \wedge T \equiv (p \vee \neg q) \wedge (\neg p \vee q)$
3. Let us try $p \equiv T$:
 $(T \vee \neg q) \wedge (\perp \vee q) \equiv q$
4. Let then be $q \equiv T$: the formula becomes true. Therefore, it is satisfiable, and a model is when p, q and r are all true.

5. Sets (6 + 9 points)

Consider the following set-theoretic equality:

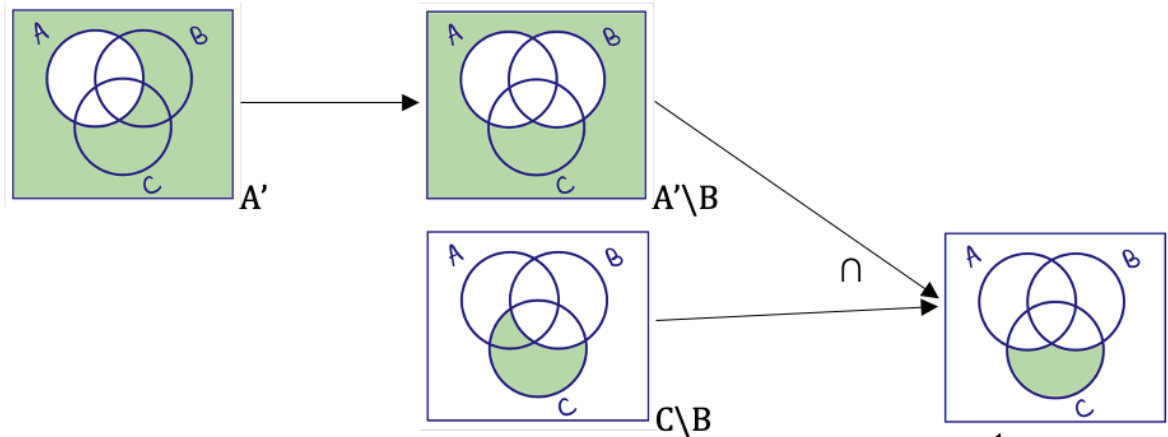
$$(A' \setminus B) \cap (C \setminus B) = (A \cup C')' \setminus B$$

- (a) Prove this identity using a sequence of Venn diagrams. For each side of the equality, you can draw a sequence of diagrams which shows how the set is constructed from A, B, and C via elementary set operations, and use this to prove that the equation holds for any three sets A, B and C.
- (b) Prove this identity using the rules of the algebra of sets. Please make sure to name the rules which you apply.

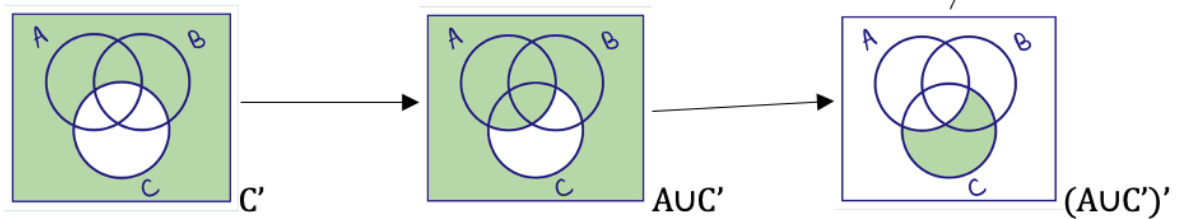
SOLUTION

(a)

Venn-diagrams for left-hand side:



Venn-diagrams for the right-hand side:



(b)

Left-hand side:

$$\begin{aligned}
 (A' \setminus B) \cap (C \setminus B) &= (A' \cap B') \cap (C \cap B') && \text{Definition of difference} \\
 &= A' \cap B' \cap C \cap B' && \text{Associativity} \\
 &= A' \cap C \cap B' \cap B' && \text{Commutativity} \\
 &= A' \cap C \cap (B' \cap B') && \text{Associativity} \\
 &= A' \cap C \cap B' && \text{Idempotence} \\
 &= A' \cap B' \cap C && \text{Commutativity}
 \end{aligned}$$

Right-hand side:

$$\begin{aligned}
 (A \cup C') \setminus B &= (A' \cap (C')') \setminus B && \text{De Morgan rule} \\
 &= (A' \cap C) \setminus B && \text{Involution} \\
 &= (A' \cap C) \cap B' && \text{Definition of difference} \\
 &= A' \cap C \cap B' && \text{Associativity} \\
 &= A' \cap B' \cap C && \text{Commutativity}
 \end{aligned}$$

The derivations above show that both sides are equivalent to $A' \cap B' \cap C$, and so equivalent to each other.

6. Relations (5 + 6 + 6 points)

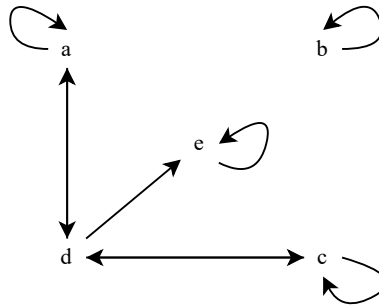
Consider the set $V = \{a, b, c, d, e\}$ and $W = \{1, 2, 3, 4, 5\}$ and the relations:

$$\begin{aligned}
 R &= \{(a, a), (a, d), (b, b), (c, c), (c, d), (d, a), (d, c), (d, e), (e, e)\} \\
 S &= \{(a, 2), (b, 3), (d, 3), (d, 5), (e, 4)\}
 \end{aligned}$$

- Draw the directed graph representation of the relation R .
- Explain whether the relation R is transitive and/or anti-symmetric.
- Write down the matrix representation of $S \circ R$. It might help to illustrate the relations in a Venn-diagram first.

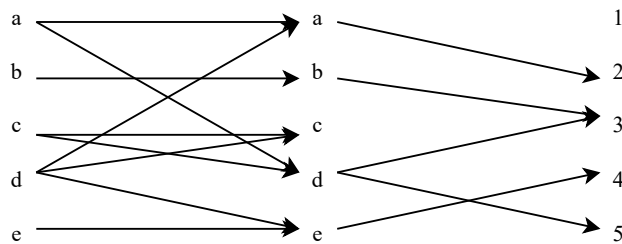
SOLUTION:

(a)



- (b) It is not transitive as it has (c, d) and (d, e) but not (c, e) . It is also not anti-symmetric as we have (a, d) and (d, a) as well as (c, d) and (d, c) .

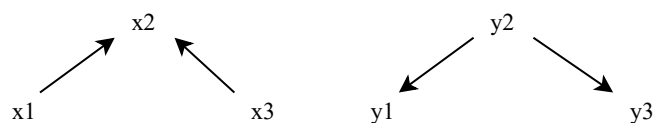
(c)



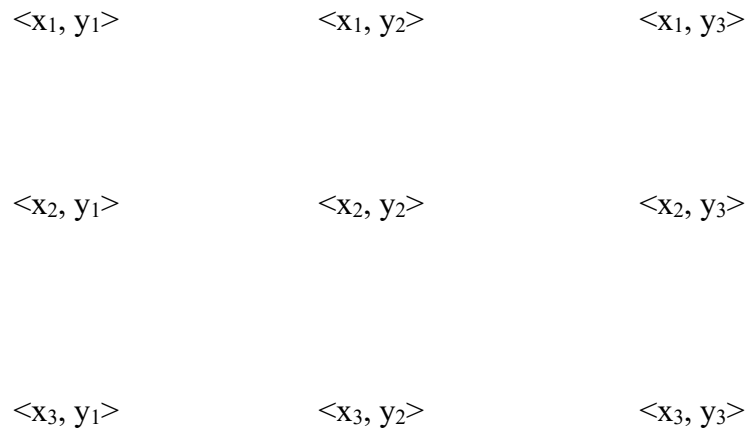
	1	2	3	4	5
a	0	1	1	0	1
b	0	0	1	0	0
c	0	0	1	0	1
d	0	1	0	1	0
e	0	0	0	1	0

7. Ordering relations (7 + 4 points)

Consider the sets $X := \{x_1, x_2, x_3\}$ and $Y := \{y_1, y_2, y_3\}$, with partial orders $\leq_X$ and $\leq_Y$ on these respective sets described by the following Hasse diagrams:



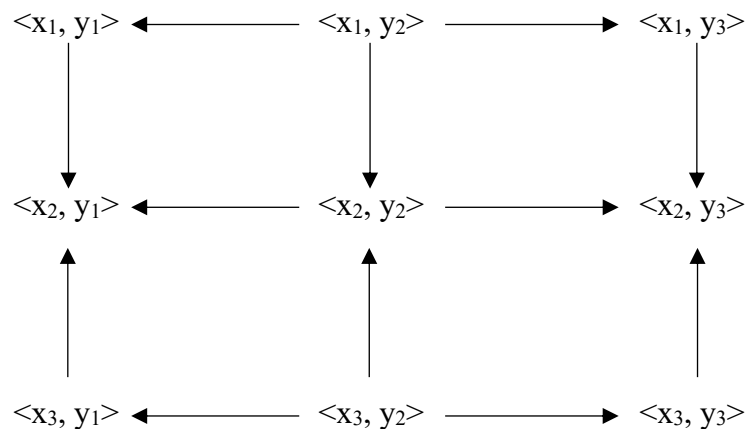
- (a) Draw the Hasse diagram of the Cartesian ordering relation on $X \times Y$ induced by $\leq_X$ and $\leq_Y$ by adding arrows to the picture below. You can use the algorithm you have learned about in class, but you are not required to do so.



- (b) Determine and list all the maximal and all the minimal elements of $X \times Y$ in this Cartesian ordering. Is there a maximum? Is there a minimum?

SOLUTION

(a)



- (b) There is not a maximum, but there are two maximal elements, $\langle x_2, y_1 \rangle$ and $\langle x_2, y_3 \rangle$. Similarly, there is not a minimum but there are two minimal elements, $\langle x_1, y_2 \rangle$ and $\langle x_3, y_2 \rangle$.